

PEDRO BARBOSA

Fullstack AI Engineer — TypeScript, Python, APIs and Automation

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Portuguese (native) · English (fluent) · German (basic)

SUMMARY

Developer who builds and ships production software and works with AI tools every day, not occasionally. I build automated workflows with n8n and Zapier, and my development workflow runs on coding agents — Claude and similar tools — driving multi-step tasks while I review every change and keep architecture decisions with me. On the engineering side I integrate systems through REST APIs and webhooks, model relational data, and own what I build: implementation, tests, quality and ongoing support. Curious by default: I taught myself Go and shipped a 5,000-line daemon with more test code than production code.

EXPERIENCE

Full Stack Developer — TIIX (remote)

April 2026 – present

- One of two developers on a healthcare platform in production, used daily by patients, clinics and professionals — the same health and benefits space this role targets.
- System integration and automation: REST endpoints, third-party APIs (maps, payments, messaging), webhooks, and automated workflows built with n8n and Zapier, with failure handling and reprocessing.
- Own the full lifecycle of what I deliver: relational modelling and versioned migrations (23 of the repository's 50), implementation, automated tests, and support after release.
- Ship features end to end, including an in-app social feed, an emergency module and the product's multi-language layer.
- Use AI development tools daily: agent-driven multi-step tasks, reviewing every generated diff, prompt iteration, and context management across large codebases.

Software Engineer — QuantConnect (freelance, remote)

September 2025 – present

- Python pipelines and machine learning models over large time-series datasets; automated batch runs and live execution against external broker APIs.
- Data engineering fundamentals in practice: ingestion, cleaning, alignment across sources, and detection of anomalies that silently corrupt results.

Python Developer — Freelance via Workana (remote)

January 2025 – March 2026

- Built MVPs directly with clients: requirements, proposal, delivery and follow-up — Django systems with role-based access and normalised relational schemas.
- Replaced manual processes with automated pipelines (pandas, openpyxl) and built dashboards on top of them.

Software Development Intern — TT&T Soluções em Informática (remote)

August 2024 – January 2025

- Django backend with payment gateway integration; MySQL modelling; unit tests with pytest; query and request optimisation.

PROJECTS

iode — open source daemon in Go (sole author). Orchestrates external tools as subprocesses under a versioned contract: bounded timeouts, process-group termination, filtered environments, and validation of every response before it is trusted; job queue with backoff and orphan recovery. The same problems agent-to-tool architectures face. 5,100 lines of code covered by 5,675 lines of tests.

Interactive 3D portfolio — React, TypeScript and Three.js: custom shaders, scroll-driven scenes, 90+ Lighthouse performance, tuned for mid-range Android devices.

QuantStrata — Python risk engine with exact arithmetic that refuses to output a metric it cannot support; 70 tests in a containerised CI pipeline.

Akaunting — merged contribution to an open source accounting platform.

SKILLS

Languages: TypeScript, JavaScript, Python, Go, PHP, SQL, Bash

AI: daily development with Claude and other coding agents (multi-step tasks, prompt iteration, context management, MCP-based tooling); academic background in machine learning (scikit-learn, pandas)

Integration and automation: n8n and Zapier (workflow automation in production use), REST APIs (design, implementation, testing), webhooks, third-party services, automated data pipelines

Data: PostgreSQL, MySQL, SQLite — modelling, queries, versioned migrations; ingestion and cleaning pipelines

Cloud and delivery: AWS EC2, Docker, GitHub Actions, GitLab CI, Linux, Git

Practice: automated testing, code review, documentation, root-cause analysis, ownership of delivery and support

EDUCATION AND CERTIFICATIONS

Computer Engineering (Data and Machine Learning) — Centro Universitário IBMR · 2023 – 2028 (in progress) · GPA 8.7/10

Machine Learning certificate and scientific initiation in Economic Mathematics — IMPA, 2024

HackerRank: Problem Solving (Intermediate), REST API (Intermediate), Python · Coursera: Python for Data Science and AI